

26/11/25

Backpropagation in CNN

$$f(x, y, z) = (x+y)z; (q = x+y)$$

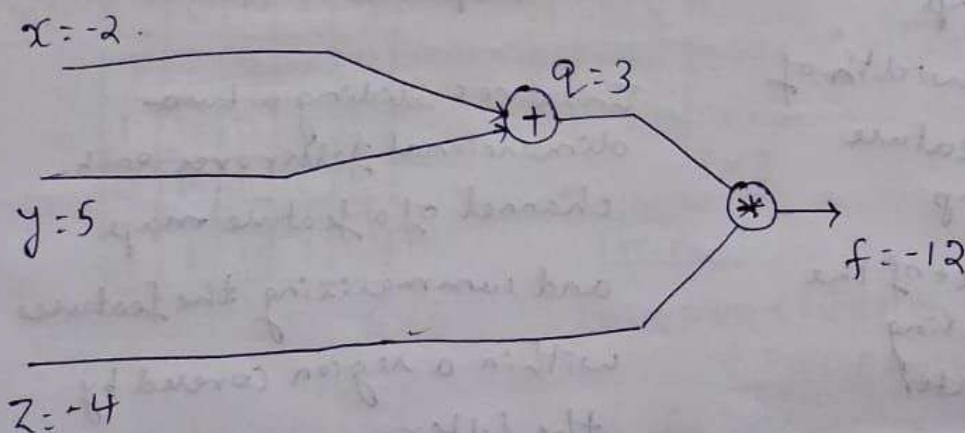
$$\text{So, } \boxed{f = qz}$$

When you differentiate 'q' with x & y

$$\frac{\partial q}{\partial x} = 1 \quad \bigg| \quad \frac{\partial q}{\partial y} = 1$$

Now let us draw the computational graph for it and its values

$$\begin{aligned} x &= -2 \\ y &= 5 \\ z &= -4 \end{aligned}$$



$$\frac{\partial f}{\partial q} = \frac{-12}{3} = -4$$

$$\frac{\partial f}{\partial z} = \frac{-12}{-4} = +3$$

$$\frac{\partial f}{\partial f} = 1$$

Now let's find

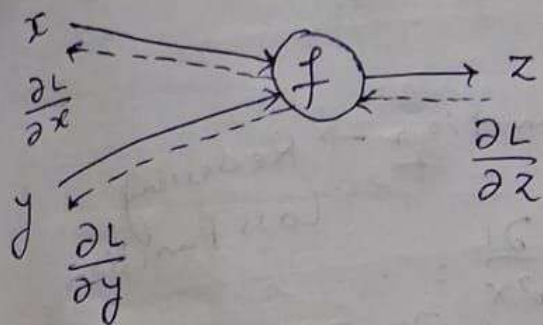
$$\frac{\partial f}{\partial x} \text{ \& \; } \frac{\partial f}{\partial y} ?$$

use chain formula

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial q} \times \frac{\partial q}{\partial x} = -4(1) = -4$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial q} \times \frac{\partial q}{\partial y} = -4(1) = -4$$

Forward propagate and then Backward propagation



So, $\frac{\partial L}{\partial x} = \frac{\partial L}{\partial z} \times \frac{\partial z}{\partial x}$

$$\frac{\partial L}{\partial y} = \frac{\partial L}{\partial z} \times \frac{\partial z}{\partial y}$$

$\frac{\partial z}{\partial x}$ & $\frac{\partial z}{\partial y}$ are ~~to~~ called as local gradients

I/p matrix 3×3 and Kernel 2×2

x ₁₁	x ₁₂	x ₁₃
x ₂₁	x ₂₂	x ₂₃
x ₃₁	x ₃₂	x ₃₃

Input

f ₁₁	f ₁₂
f ₂₁	f ₂₂

~~Output~~
Kernel/
Filter

o ₁₁	o ₁₂
o ₂₁	o ₂₂

Output

From the convolution layer
using Convolution operation
of Input & Filter/Kernel

$$O_{11} = X_{11} F_{11} + X_{12} F_{12} + X_{21} F_{21} + X_{22} F_{22} \quad \text{--- (1)}$$

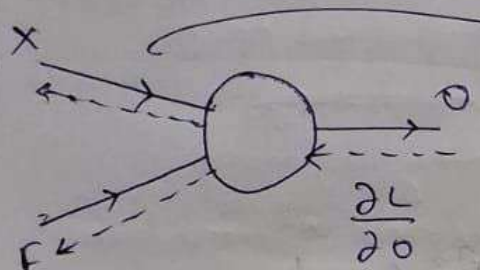
$$O_{12} = X_{12} F_{11} + X_{13} F_{12} + X_{22} F_{21} + X_{23} F_{22} \quad \text{--- (2)}$$

$$O_{21} = X_{21} F_{11} + X_{22} F_{12} + X_{31} F_{21} + X_{32} F_{22} \quad \text{--- (3)}$$

$$O_{22} = X_{22} F_{11} + X_{23} F_{12} + X_{32} F_{21} + X_{33} F_{22} \quad \text{--- (4)}$$

forward pass

Backpropagation for loss function $\rightarrow$ Reducing Loss function



$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial o} \frac{\partial o}{\partial x}$$

$$\frac{\partial L}{\partial f} = \frac{\partial L}{\partial o} \frac{\partial o}{\partial f}$$

We need to find 1st Local Gradients

$$\frac{\partial L}{\partial x} = ?, \quad \frac{\partial L}{\partial f} = ?$$

$$1) \frac{\partial L}{\partial f} = \frac{\partial L}{\partial o} \frac{\partial o}{\partial f}$$

similarly, for others
 O_{12}, O_{22}, O_{21}

$$\Rightarrow \boxed{\frac{\partial O_{11}}{\partial F_{11}} = X_{11}} \quad (\text{from (1)})$$

$$\boxed{\frac{\partial O_{11}}{\partial F_{21}} = X_{21}} \quad (\text{from (1)})$$

$$2) \boxed{\frac{\partial O_{12}}{\partial F_{12}} = X_{12}} \quad (\text{from (2)})$$

$$\boxed{\frac{\partial O_{11}}{\partial F_{22}} = X_{22}} \quad (\text{from (1)})$$

Now using chain Rule, we need to find $\frac{\partial L}{\partial F}$

$$\frac{\partial L}{\partial F} = \frac{\partial L}{\partial O} \times \frac{\partial O}{\partial F} \rightarrow \text{local}$$

$\downarrow$
loss

For every element 'i', F_i

$$\frac{\partial L}{\partial F_i} = \sum_{k=1}^m \frac{\partial L}{\partial O_k} \cdot \frac{\partial O_k}{\partial F_i}$$

Expanding we get

$$\frac{\partial L}{\partial F_{11}} = \frac{\partial L}{\partial O_{11}} \frac{\partial O_{11}}{\partial F_{11}} + \frac{\partial L}{\partial O_{12}} \frac{\partial O_{12}}{\partial F_{11}} + \frac{\partial L}{\partial O_{21}} \frac{\partial O_{21}}{\partial F_{11}} + \frac{\partial L}{\partial O_{22}} \frac{\partial O_{22}}{\partial F_{11}}$$

similarly

$$\frac{\partial L}{\partial F_{12}}, \frac{\partial L}{\partial F_{21}}, \frac{\partial L}{\partial F_{22}}$$

$$\text{So, } \frac{\partial L}{\partial F_{11}} = \frac{\partial L}{\partial O_{11}} x_{11} + \frac{\partial L}{\partial O_{12}} x_{12} + \frac{\partial L}{\partial O_{21}} x_{21}$$

$$\Downarrow + \frac{\partial L}{\partial O_{22}} x_{22}$$

$$\frac{\partial L}{\partial F_{11}} = \text{conv} \begin{pmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{pmatrix} \begin{pmatrix} \frac{\partial L}{\partial o_{11}} & \frac{\partial L}{\partial o_{12}} \\ \frac{\partial L}{\partial o_{21}} & \frac{\partial L}{\partial o_{22}} \end{pmatrix}$$

3×3 2×2

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial o} \frac{\partial o}{\partial x}, \quad \text{1. finding local gradient} \quad = - \frac{\partial o}{\partial x}$$

$$\boxed{\frac{\partial o_{11}}{\partial F_{11}} = x_{11}, \quad \frac{\partial o_{11}}{\partial F_{12}} = x_{12}, \quad \frac{\partial o_{21}}{\partial F_{21}} = x_{21}, \quad \frac{\partial o_{21}}{\partial F_{22}} = x_{22}}$$

(from equation ①)

Similarly, ~~$\frac{\partial o_{12}}{\partial F_{11}}$~~ , for o_{12}, o_{22}, o_{21}

$$\frac{\partial L}{\partial x_i} = \sum_{k=1}^m \frac{\partial L}{\partial o_k} \frac{\partial o_k}{\partial x_i}$$

Expanding

$$\boxed{\frac{\partial L}{\partial x_{11}} = \frac{\partial L}{\partial o_{11}} f_{11}} + \boxed{\frac{\partial L}{\partial o_{12}} f_{12} = \frac{\partial L}{\partial x_{12}}}$$